
COORDINATION WITHOUT COMMITMENT: FAILURE MODES OF LLM AGENTS ON DISTRIBUTED COMPUTING TASKS AND A BELIEF-BASED REMEDY

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ABSTRACT

Multi-agent systems of large language models (LLMs) demonstrate strong capability on individual reasoning tasks, yet coordination accuracy degrades sharply as communication graphs grow in size. We study this degradation on five canonical distributed-computing tasks — consensus, leader election, matching, graph coloring, and vertex cover — using the AgentsNet synchronous message-passing benchmark, and report hard-pass scores (fraction of fully solved instances) as the primary metric, with soft scores used only as diagnostics. Evaluating two open-weight models (qwen3.5-4b and qwen3.5-9b), we find that coordination failures are not tied to single failure mode but have distinct dynamic failure reasons. We further show that the apparent strength for the consensus task is an artefact of inductive bias of the underlying language model. With explicit state initialisation, the focal-point head start disappears and consensus drops from 0.93 to 0.65 for a model with 4 billion parameters. To address the perception-limited failures, we introduce the **Neighbour Belief Protocol (NBP)**, a protocol layer in which each agent maintains and broadcasts a typed belief about its immediate neighbours on every message. On perception-bottlenecked tasks, NBP lets models with as few as 4 billion parameters reach or exceed the next model scale and close roughly 60% (Global) and 48% (Local) of the hard-pass gap to the best frontier models. Coloring, whose bottleneck is a structural synchrony problem rather than about missing information, is not repaired by NBP and is made marginally worse. Our results suggest that targeted belief scaffolding can substitute for scale on perception-limited tasks, but not on structurally limited ones. Fundamentally perception, local reasoning, and information scope remain separable bottlenecks.

1 INTRODUCTION

Large Language Model (LLM) agents are increasingly deployed not in isolation but in multi-agent systems, where collections of agents collaborate to solve tasks that exceed the capacity or context of a single model (Guo et al., 2024; Chen et al., 2025). There are several structural patterns for designing and deploying such a system. Sequential frameworks have chains of agents wherein each agent processes the output of the previous one. Centralised systems have a single controller that dispatches subtasks and aggregate results. Frameworks such as AutoGen (Wu et al., 2023), Crew AI follow this centralised or hierarchical pattern. A structurally different class is *decentralized* multi-agent systems, in which each agent holds only local information and communicates solely with its immediate neighbours in a communication graph.

As LLM agents are deployed jointly across organizational boundaries for scientific research, cross-enterprise supply chains or emergency-response networks, no single organization can act as a trusted central controller (Yao et al., 2025; Ko et al., 2025). In the future, swarms of agents from different providers are expected to form ad-hoc coalitions and solve large tasks through local exchange alone, without surrendering proprietary data or submitting to a third-party orchestrator (Ko et al., 2025). Decentralized coordination is also inherently more fault-tolerant and scalable beyond a single bottleneck node (Yao et al., 2025).

Despite these motivations, coordination in decentralized LLM multi-agent systems is less studied than centralized pipelines. Recent work has begun to examine decentralized agent networks from complementary angles. AgentNet motivates decentralization as a way to reduce centralized bottlenecks and single points of failure (Yao et al., 2025), SwarmBench evaluates LLM agents under local perception and local communication constraints and finds that current models still struggle with robust planning under decentralized uncertainty (Ruan et al., 2025) and AgentsNet shows that performance on distributed-computing coordination tasks generally drops as the number of agents increases (Grötschla et al., 2025). These findings motivate a closer analysis of where graph-structured LLM agents fail, and whether protocol-level changes can mitigate the resulting bottlenecks.

To investigate where such multi-agent systems break down we conduct experiments on AgentsNet (Grötschla et al., 2025), a benchmark of five classical distributed-computing tasks — consensus, leader election, matching, graph coloring, and vertex cover — evaluated on communication graphs of N nodes with synchronous peer-to-peer message passing and no central coordinator. We focus on small open-weights models for two reasons. They represent the realistic deployment frontier for edge and privacy-constrained settings where decentralized coordination is most needed (Lu et al., 2025), and the performance gap under local communication is widest at this scale (Grötschla et al., 2025), making protocol-level improvements most detectable relative to model-scale effects.

We conduct our investigation in two halves. First, we try to figure out *why* small models fall short by examining how the agents’ performance and outputs varies across tasks, considering both quantitative and qualitative aspects. Along the way we also show that the apparent ease of consensus is an evaluation artefact. Under the default task framing for consensus, agents are not assigned a state during the first round of communication. As a result, more than half of the runs begin with all agents already at the same state due to the model’s inductive bias to pick 0 over 1, and so the protocol succeeds by doing no work. We then report **hard-pass scores** (fraction of fully solved instances, following Grötschla et al., 2025) under balanced *hard* initialisation as the honest test where the initial states for each agent are assigned by the task controller.

Second, we ask whether perception-limited failures can be repaired at the protocol level. We introduce the *Neighbour Belief Protocol* (NBP), a prompt-and-schema layer addition in which each agent maintains and broadcasts a typed belief that includes inferred state and the confidence assigned to that inference. We explore two variants. In the Local variant, each agent’s belief covers only its k direct neighbours, giving k belief entries per message; we ablate this against a Global variant covering all n nodes (N entries) to isolate the effect of belief scope (§5). We find that NBP improves precisely the tasks whose bottleneck is neighbour perception namely consensus, leader election, and matching, and that these gains are most pronounced at the smaller model scale where the coordination gap is widest. On these tasks, qwen3.5-4b with NBP reaches or exceeds the next model scale and closes roughly 60% (Global) and 48% (Local) of the hard-pass gap to the best frontier models. Graph coloring remains unsolved by either variant where its bottleneck is structural rather than missing information, and adding belief coupling makes the existing oscillation marginally worse. Together, these results indicate that perception failures and local reasoning limitations are separable bottlenecks in decentralised LLM coordination. With NBP, we address the first, while the second persists regardless of belief scope.

2 RELATED WORK

Early multi-agent LLM work centred on debate where multiple model instances iteratively critique each other’s answers, reducing hallucination and improving factuality (Du et al., 2023; Liang et al., 2023). Frameworks such as CAMEL (Li et al., 2023a) and AutoGen (Wu et al., 2023) generalised this to role-playing and tool-augmented conversation pipelines. MetaGPT (Hong et al., 2023) uses SOP-driven structured artifacts such as requirements, designs, and diagrams reducing ambiguity relative to unconstrained dialogue. DebUnc (Yoffe et al., 2024) took the next step by augmenting debate messages with per-agent uncertainty estimates, finding that confidence-weighted communication improves collective accuracy. This is the closest prior work to NBP’s typed {STATE, CONFIDENCE} messages, though in a centralised fully-connected debate setting rather than a graph-structured peer-to-peer one.

AgentBench (Liu et al., 2023) established evaluation of LLMs as single interactive agents; subsequent benchmarks extended this to multi-agent settings. LLM-Coordination (Agashe et al., 2025)

probed pure coordination games and found theory-of-mind reasoning to be a consistent bottleneck. MAgIC (Xu et al., 2023) added social-deduction and game-theoretic scenarios, while MultiAgentBench (Zhu et al., 2025) compares star, chain, tree, and graph communication protocols, finding graph-based coordination strongest in its scenarios. SwarmBench (Ruan et al., 2025) SwarmBench highlights failures under local perception and decentralized uncertainty. AgentsNet (Grötschla et al., 2025) grounds evaluation in five classical distributed-computing tasks with strict correctness criteria and shows sharp performance degradation as graph size grows. The AgentsNet benchmark is particularly suited here because protocol changes can be compared quantitatively against known baselines.

Kosinski (2023) reported emergent false-belief reasoning in large LLMs, but ToMBench (Chen et al., 2024) showed systematic gaps remain at scale. Li et al. (2023b) demonstrated in a cooperative text game that explicit natural-language belief representations outperform implicit inference which is the direct empirical motivation for our NBP design. Concurrently, Wang et al. (2025) argues that existing LLM collaboration frameworks underuse dynamic intent inference, leading to inconsistent plans and redundant communication. We address the same bottleneck in a decentralised setting through per-neighbour typed beliefs rather than a shared global structure.

The tasks in AgentsNet are drawn from the classical distributed-computing literature. Consensus under asynchrony is provably impossible with even one crash failure (Fischer et al., 1985) and randomised protocols circumvent this (Ben-Or, 1983). Gossip algorithms (Demers et al., 1987) achieve information dissemination through unstructured local exchange. Distributed-locality theory (Linial, 1992) separates tasks solvable with $O(1)$ -hop information from those requiring $\Omega(\log^* n)$ or $\Omega(D)$ rounds. NBP is structurally a *structured gossip* protocol: the typed `{STATE, CONFIDENCE}` belief messages are analogous to belief propagation in spirit to (Pearl, 1988), in which nodes maintain and exchange estimates over a graph, but the updates are prompt-level natural-language beliefs instead of formal probabilistic messages.

3 THE MULTI-AGENT COORDINATION PROBLEM

The AgentsNet benchmark (Grötschla et al., 2025) places LLM agents at the nodes of a random graph and asks them to solve five canonical distributed-computing problems through synchronous message passing for a fixed number of message passing rounds. Each vertex is an LLM agent with separate conversation memory. At round $t = 0$, before any messages have been received, each agent observes only its task instructions, own identity, neighbour list, and any task-specific initial state, and emits its first message to its neighbours. In round $t > 0$, agent v : (1) reads the messages its neighbours sent in round $t-1$; (2) updates its private `state` field in a structured JSON response; and (3) emits one outgoing message per neighbour. After R rounds, each agent produces a final answer. Following AgentsNet, we set the number of rounds to $R = 2 \cdot \text{diameter}(G) + 1$ for the consensus and leader election tasks, and set it explicitly for other tasks mimicing the original benchmark. There is no single central coordinator.

In Table 1, we list the 5 tasks in the benchmark along with the goal of the network for each of these tasks and what state means per round per task. Figure 1 shows one solved and one failed instance of each, illustrating what success and failure look like. Node colour shows each agent’s committed `state` and red edges mark constraint violations.

Task	Objective	state
Consensus	Agree on one binary value	tentative value, 0/1
Leader election	Elect exactly one leader	self-leader status, Yes/No
Coloring	Give neighbours different groups	chosen group/color
Matching	Form a maximal set of consistent neighbour pairs	proposed partner or None
Vertex cover	Select a minimal set of coordinators covering every edge	coordinator status, Yes/No

Table 1: AgentsNet tasks and the meaning of the per-round `state` field.

Evaluation metric. AgentsNet’s primary metric is **hard pass** i.e., an instance counts only if the entire networks converges to satisfy the task objective under the given constraints (score = 1.0).

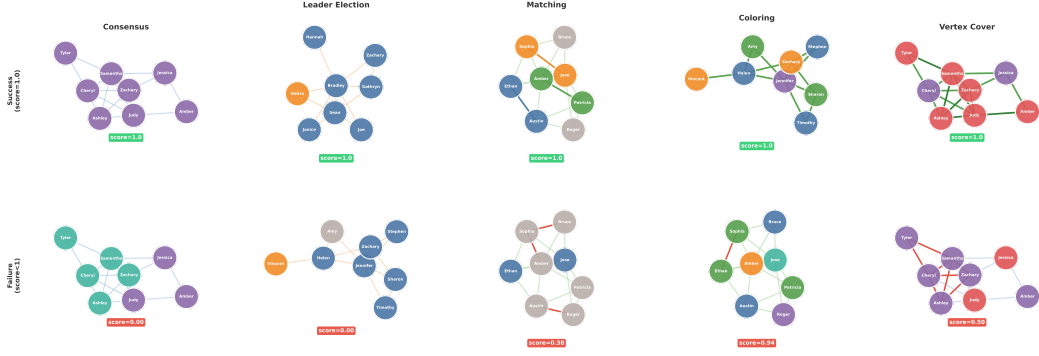


Figure 1: Task instances at $n=8$ from the qwen3.5-4b corpus. Top row: fully solved (hard pass, score = 1.0). Bottom row: representative failures, with red edges marking violations like non-reciprocal matches, an illegal same-colour edge, and an uncovered edge. Node colour encodes each agent’s committed state; node label is the agent name. Note: the scores in the bottom row are soft scores which indicate partial success.

Consensus and leader election are inherently binary meaning the result can be only right or wrong; matching, coloring, and vertex cover return fractional soft scores, but we aggregate the fraction of fully solved instances for all headline comparisons, following Grötschla et al. (2025). Soft mean scores are reported only where partial progress is diagnostically useful.

3.1 CONSENSUS: APPARENT VERSUS REAL PERFORMANCE

Consensus appears to be the easiest task in the default AgentsNet benchmark, with both 4B and 9B models achieving success rates close to 0.90. However, this performance is largely driven by an artifact of the task design. In the original benchmark, agents are not assigned an initial state; instead, each agent must independently choose a value constrained to either 0 or 1. In practice, we found that the agents exhibit a strong prior toward 0, causing a large fraction of agents to select the same value even before any communication takes place. As shown in Figure 2, more than half of the default-initialization runs begin with nearly all agents selecting 0. In these cases, consensus is effectively achieved at initialization, requiring little or no coordination during protocol execution.

To remove this bias, we modify the task by explicitly assigning initial states to agents. We introduce a parameter p that controls the proportion of agents initialized to ‘0’ versus ‘1’, allowing us to systematically vary the difficulty of the consensus problem. This produces a controlled distribution of initial disagreement and ensures that successful consensus requires coordination rather than reliance on a shared default choice.

Hard initialisation ($p=0.5$): each agent starts at ‘0’ or ‘1’ with equal probability removing the gain from inherent bias and matches the random-init protocol used for the NBP cohorts in our main tables. As shown in Figure 2, under the hard initialisation setting, the performance of the 4B model drops from ~ 0.93 to ~ 0.65 in soft score, while the performance of the 9B model drops from ~ 0.89 to ~ 0.80 showing that the headline number was mostly driven by inductive bias of the language models to default to 0. All primary consensus results in this paper are reported under the hard initialisation setting, unless explicitly labelled as “easy init.”

4 FAILURE ANALYSIS

Across two open-weight models (qwen3.5-4b and qwen3.5-9b), three graph sizes ($n \in \{4, 8, 16\}$), and three graph structure generators (Watts–Strogatz, Barabási–Albert, Delaunay) with three samples per cell, we try to answer *what behaviour characterises a failed run?* The only signal every task exposes is each agent’s tentative state per round, so we begin with two state-only diagnostics (reported here for qwen3.5-4b) and then parse the message content itself on the cleanest task, consensus.

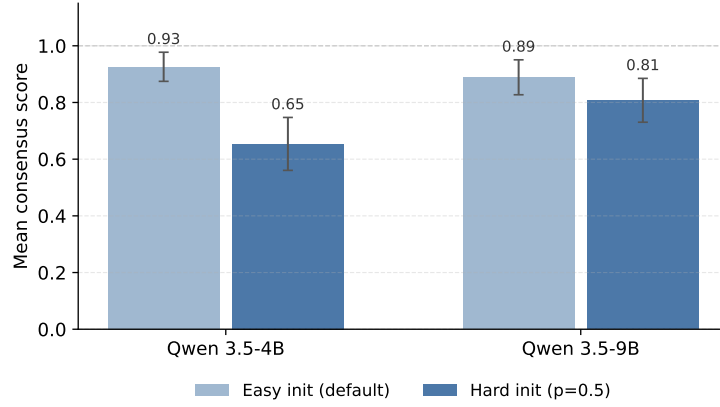


Figure 2: Consensus soft score under easy (default) versus hard ($p=0.5$) initialisation. The agents lose most of the apparent performance once the focal-point head start is removed. 52% of easy-init runs on the 4B parameter harness begin with all agents already at '0'. SEM error bars.

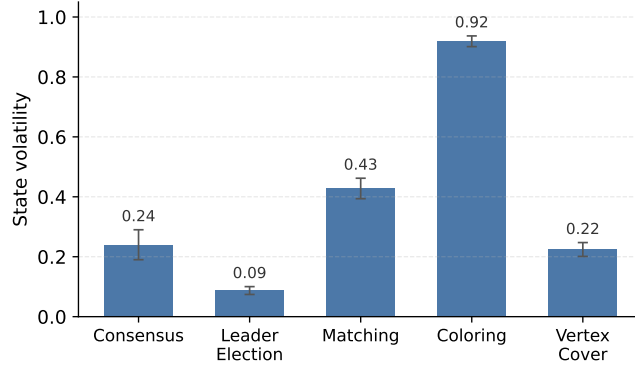


Figure 3: State volatility (fraction of consecutive rounds in which an agent changes its tentative state) by task for `qwen3.5-4b`; SEM error bars. Coloring agents thrash continuously while leader-election agents freeze.

State volatility. We define state volatility as, for each agent, the fraction of consecutive rounds in which its declared `state` changes, averaged over agents and runs. High volatility indicates that agents repeatedly revise their tentative commitments rather than settling. The two extremes in Figure 3 are the most informative: coloring is near-maximal (0.92), with agents shifting groups in lockstep and rarely stabilising, whereas leader election is near-zero (0.09), consistent with agents locking into stable candidacy decisions early rather than breaking symmetry. Consensus and vertex cover are only mildly volatile, suggesting that their failures are less about continual state thrashing and more about what information agents use to settle.

Final-answer mismatch. We then compare each agent’s committed `final answer` against the tentative `state` it declared in its *last* message round where mismatch is the fraction of agents for which the two disagree. This diagnostic asks whether the final-answer prompt reads out the agent’s last deliberative commitment or induces a late revision after receiving the final neighbor messages. As Figure 4 shows, coloring has a high mismatch rate ($\sim 70\%$), suggesting that agents often change commitments at the final query rather than carrying forward a settled state. Consensus and leader election remain low, consistent with their comparatively stable final-round states. High mismatch therefore marks an unstable decision interface: extra rounds may help only if they reduce late switching, not merely by adding more rounds for deliberation.

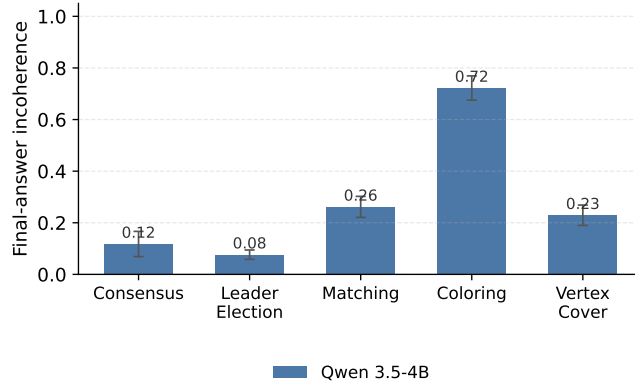


Figure 4: Final-answer incoherence: fraction of agents whose committed answer differs from the tentative state they declared in their last message round (qwen3.5-4b, SEM error bars). In coloring and matching the final commit is largely decoupled from the agent’s own last broadcast.

Probing consensus: belief versus reality. The diagnostics above localise that some networks fail to settle. To see the mechanism behind this, we look at consensus as states are binary (0/1) and the target is global agreement. Parsing each agent’s outgoing messages, we score three kinds of claim against the ground-truth states: a self-claim (“my value is X”), a claim about a neighbour (“you are at X”), and an agreement claim (“consensus reached”, “finalising on X”). The result (Figure 5) is a sharp asymmetry where agents report their *own* state with 98% accuracy but predict a *neighbour’s* state at only 63% (chance is 50%), and 47% of all agreement claims are issued while the network is in fact split. (A fuller catalogue of each claim-failure type, and task transcript walk-throughs is deferred to Appendix A.)

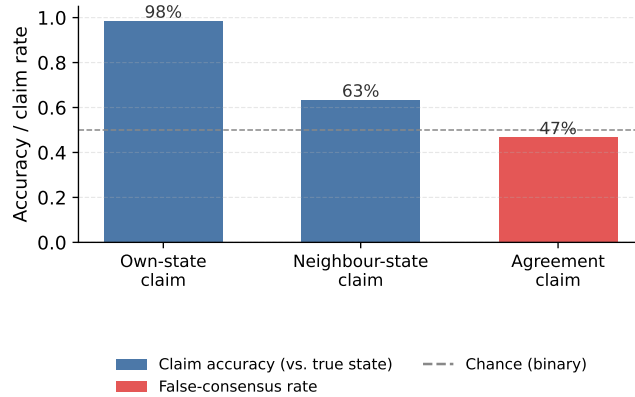


Figure 5: Consensus claim accuracy under hard initialisation ($p=0.5$, qwen3.5-4b). Agents model their own state almost perfectly (98%) but a neighbour’s barely above chance (63% vs. 50%); nearly half of explicit agreement claims (47%) are made while the network is split. The failure is in modelling others’ state, and it surfaces as false consensus.

We see that the default communication protocol gives each agent a slot for its own state and free-text messages, but no persistent, structured representation of what its neighbours are doing. Misperceptions are never surfaced or corrected leading to early incorrect commitment. This directly motivates the protocol change we introduce next.

Table 2: Protocol cohorts. Each row adds one mechanism. k is node degree, n is graph size.

Cohort	state field	belief vector	payload/msg
A. Structured Baseline	per agent	–	–
B. Global NBP	per agent	$O(n)$	$\Theta(n)$
C. Local NBP	per agent	$O(k)$	$\Theta(k)$

5 THE NEIGHBOUR BELIEF PROTOCOL

We hypothesise that giving each agent an explicit, persistent record of its neighbours’ states and forcing it to broadcast that record can recover the perception-limited failures without scaling the model.

Belief vector. Agent v maintains a dictionary b_v keyed by v and its neighbours $N(v)$. Each entry $b_v[u] = (\text{STATE}_u, \text{CONF}_u)$ where $\text{CONF} \in \{\text{HIGH}, \text{MEDIUM}, \text{LOW}, \text{UNKNOWN}\}$: HIGH for directly observed states, MEDIUM for one-hop relayed beliefs, LOW for stale or two-hop information, and UNKNOWN otherwise. The vector has $|N(v)|+1$ entries; since average degree in our graphs is nearly constant in n , the per-message belief payload stays small as graphs grow.

Message format. Each per-round message is JSON:

```
{
  "state":           "<commitment>",
  "belief_state":    { ...v's belief vector... },
  "messages":        { "u": "<text>" }
}
```

where `state` is private to v , `belief_state` is relayed, and `messages` holds per-neighbour text. On receipt, the we remove the recipient’s own entry from the sender’s belief before passing it on. Agents update b_v from observed neighbour states and relayed beliefs. No task-specific update rule is hard-coded.

Variants. *Local NBP* keys the belief by the agent’s immediate 1-hop neighbours (at most $\max_k$ entries per message); it is cheaper, more scalable, and free of the aggregation errors that accumulate over large networks. *Global NBP* keys the belief by every node ($O(n)$ entries, $\Theta(|E| \cdot n)$ traffic per round) and serves as an upper-bound reference on information scope. We compare both against structured-JSON baseline protocol (Table 2).

6 EXPERIMENTS

We evaluate the protocol using Qwen3.5 over two model sizes (4B and 9B) served locally via vLLM (Kwon et al., 2023). Graphs come from the public HuggingFace dataset `disco-eth/AgentsNet` (Grötschla et al., 2025) at $n \in \{4, 8, 16\}$, three samples per cell (27 runs per task per cohort). For consensus, each agent’s initial state is drawn i.i.d. Bernoulli(0.5) for cohorts B–D (hard init). Implementation and reproducibility details are in Appendices D and E.

6.1 WHERE THE BELIEF PROTOCOL WORKS

Figure 6 and Table 3 report hard-pass scores across all tasks, protocols, and both model scales.

Perception-bottlenecked tasks. On **consensus**, hard pass rises from 0.63 (Structured) to 0.89 (Global) / 0.85 (Local) on 4B: the explicit belief record converts focal-point-seeking into genuine coordination under hard initialisation. On **leader election**, both variants help but Global dominates ($0.56 \rightarrow 0.85$ Global, $\rightarrow 0.70$ Local on 4B); the belief anchor counteracts early lock-in and for symmetry-breaking it benefits from wider scope (§B). On **matching**, hard pass more than doubles, from 0.22 to 0.48 (Local) / 0.44 (Global) on 4B, with Local slightly ahead consistent with pair-local scope being sufficient when the answer depends on immediate neighbours. On all three, Local NBP tracks Global closely, indicating that $O(k)$ scope captures most of the benefit at a fraction of the message cost.

Table 3: Hard-pass scores (fraction fully solved) by task, model, and protocol; mean soft scores in gray parentheses). 27 graphs per cell at $n \in \{4, 8, 16\}$. Consensus uses hard init ($p=0.5$). **Bold** marks the best hard-pass rate within each (task, model) block.

Model	Protocol	Consensus	Leader el.	Matching	Coloring	Vertex cov.
4B	Structured	0.63 (0.65)	0.56 (0.56)	0.22 (0.69)	0.04 (0.70)	0.11 (0.72)
	Global NBP	0.89 (0.89)	0.85 (0.88)	0.44 (0.84)	0.07 (0.70)	0.19 (0.79)
	Local NBP	0.85 (0.88)	0.70 (0.73)	0.48 (0.85)	0.04 (0.56)	0.22 (0.79)
9B	Structured	0.78 (0.81)	0.85 (0.92)	0.44 (0.84)	0.41 (0.89)	0.19 (0.77)
	Global NBP	0.89 (0.92)	0.89 (0.89)	0.44 (0.84)	0.33 (0.89)	0.41 (0.85)
	Local NBP	0.93 (0.93)	0.93 (0.96)	0.44 (0.85)	0.31 (0.84)	0.26 (0.79)

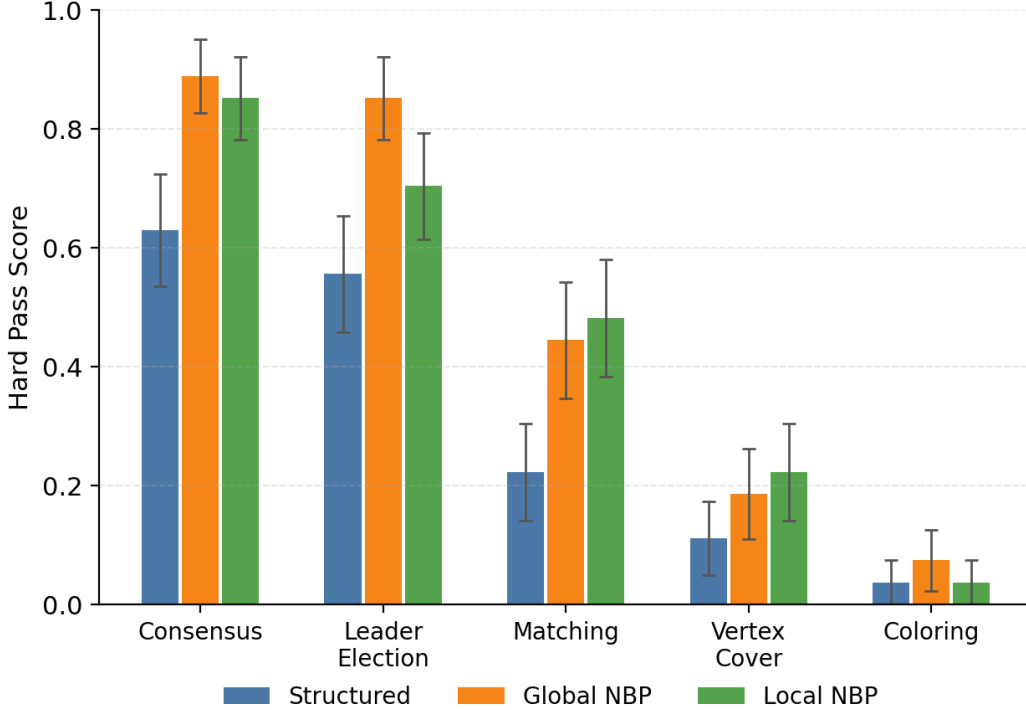


Figure 6: Hard pass (fraction of fully solved instances) by task and protocol. NBP variants (Global, Local) lift the perception-bottlenecked tasks (consensus, leader election, matching) at 4B by a significant margin, where the coordination gap is widest. Coloring shows no consistent benefit.

Why it works: the belief records are calibrated. These gains rest on the belief field carrying real information rather than boilerplate. We group every belief entry an agent emits by its self-reported confidence and score the asserted state against the ground-truth `round_states` of that round (Figure 7). Accuracy tracks confidence monotonically: a high-confidence belief is correct about 75% of the time, falling to $\sim 10\%$ at low and $\sim 2\%$ at unknown, where the agent effectively declines to assert. The confidence labels are meaningful whereby agents estimate accurately what they know. This lets the belief record serve as an commitment anchor.

6.2 WHERE NBP DOESN'T WORK: COLORING IS STRUCTURALLY HARD

On 4B, hard pass stays at the floor under every protocol (≤ 0.07), and Local NBP actually *lowers* the soft score relative to the baseline. The mechanism is visible in the lockstep statistics (Figure 8): the mean fraction of round-to-round transitions in which $\geq 70\%$ of agents simultaneously flip rises

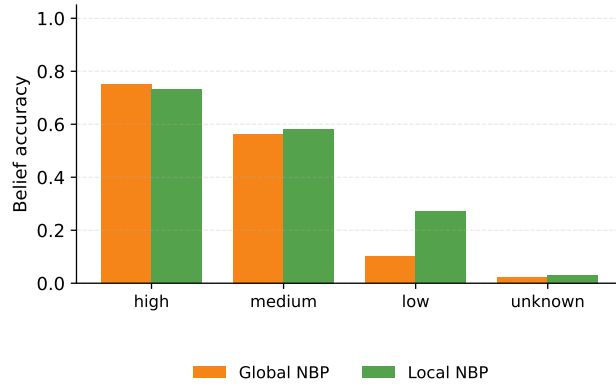


Figure 7: Belief accuracy versus self-reported confidence under NBP (qwen3.5-4b, pooled over all tasks). Accuracy decreases monotonically from high to unknown for both scopes, so the confidence label predicts state and useful calibrated confidence labels.

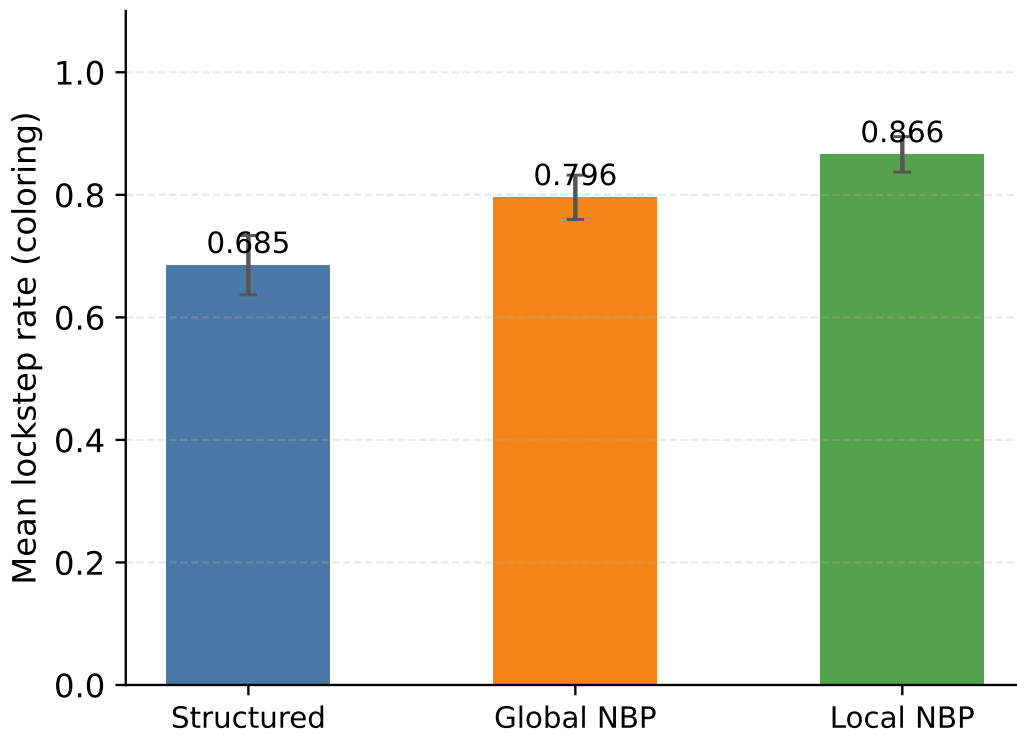


Figure 8: Coloring lockstep. Mean lockstep rate by protocol. Adding belief coupling shifts mass toward total lockstep (1.0) making the network more unstable.

from 0.685 (Structured) to 0.796 (Global) to 0.866 (Local) across both model sizes. More explicit belief coupling tightens the oscillation rather than damping it. This is structural and shows that NBP is a perception remedy, and not a universal one.

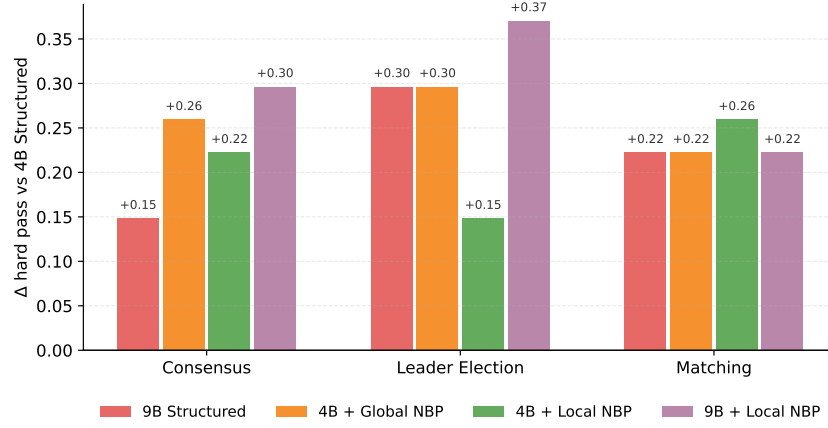


Figure 9: Hard-pass gain over the 4B Structured baseline for four interventions on the perception-bottlenecked tasks. On consensus and matching, protocol improvement (4B + NBP) matches or exceeds the gain from doubling model size (9B Structured).

6.3 PROTOCOL VERSUS SCALE: AN EFFICIENCY COMPARISON

A natural question that arises is whether it is more efficient to scale the model (4B $\rightarrow$ 9B, a $\sim 2.2\times$ increase in parameters) or to improve the protocol (Structured $\rightarrow$ NBP, a modest prompt change). In Figure 9 we try to answer this, plotting each intervention’s hard-pass gain over the 4B Structured baseline. On consensus and matching, **4B + NBP matches or exceeds 9B Structured**: 4B+Global gives +0.26 on consensus versus +0.15 for 9B Structured, and 4B+Local gives +0.26 on matching versus +0.22 for 9B. On leader election the two are comparable (+0.30 each for 9B Structured and 4B+Global). For these tasks, protocol and scale are substitutes; combining them (9B+Local NBP) pushes further still (+0.30 to +0.37).

6.4 CLOSING THE GAP TO FRONTIER MODELS

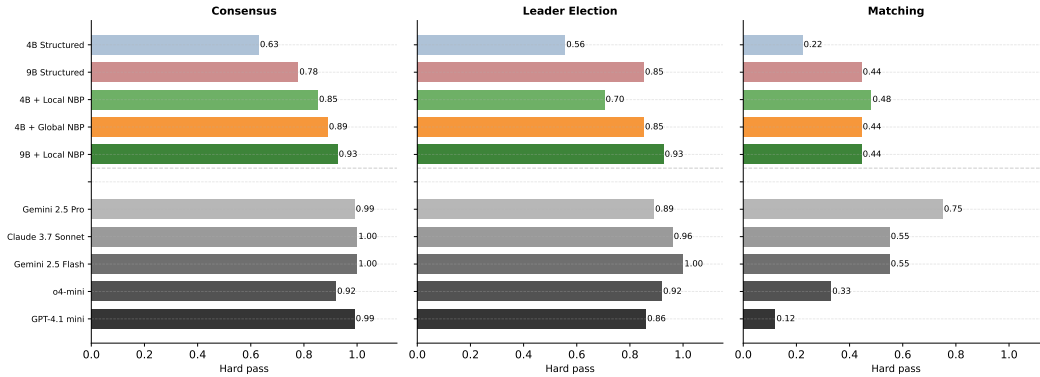


Figure 10: Hard pass on three tasks: our cohorts (coloured) versus frontier models (grey) taken from the original paper. With NBP, qwen3.5 reaches or approaches several frontier systems on consensus and leader election. Consensus rows for our cohorts use hard initialisation which means our intervention performs better even for the harder version of the task.

Figure 10 places our cohorts next to frontier models from Grötschla et al. (2025, Table 2) on hard pass. On **consensus**, 9B+Local NBP reaches 0.93, within reach of o4-mini (0.92) and trailing the strongest frontier systems (0.99–1.00). For consensus, it must be noted that their consensus task had no forced-initialisation and hence easier. On **leader election**, 9B+Local NBP reaches 0.93,

beating Gemini 2.5 Pro (0.89) and matching o4-mini (0.92). On **matching**, 4B+Local NBP reaches 0.48, surpassing GPT-4.1 mini (0.12) and o4-mini (0.33) but remaining below Claude 3.7 Sonnet and Gemini 2.5 Flash (0.55). Aggregating over the three perception tasks, 4B+Global NBP closes roughly 60% of the hard-pass gap between 4B Structured and the best frontier model, and 4B+Local NBP closes $\sim 48\%$ with small open-weight models. We note the consensus comparison mixes our hard-initialisation runs with published frontier rows that used default initialisation (inherently easier). The leader-election and matching comparisons are like-for-like.

7 CONCLUSION

We probe why small LLMs fail to coordinate on decentralised distributed-computing tasks, and whether the failures can be repaired at the communication structure level. We show that the apparent ease of consensus is an evaluation artefact of focal-point initialisation, and that hard pass under hard initialisation is the correct test.

Three of the five tasks (consensus, leader election, matching) are perception-bottlenecked, and the Neighbour Belief Protocol regains performance on these by broadcasting per-neighbour beliefs which gives agents a commitment anchor by providing a broader view of the state of the network. On these tasks NBP lets a 4B model match or exceed a 9B model and close roughly half the hard-pass gap to frontier systems, with Local ($O(k)$) scope capturing most of the benefit. Coloring is more reasoning-bottlenecked which NBP does not fix. Perception, local reasoning, and information scope are therefore separable bottlenecks and recoverable neighbour structure substitutes for scale on the first, but not on the rest.

LIMITATIONS AND FUTURE WORK

We evaluate two scales of one open model family only. This makes the scale comparison controlled, but it does not establish whether the failures and NBP gains across other models or frontier systems. Future work must focus on repeating the protocol across a broader model suite and include the evaluation of the latest frontier models directly.

The benchmark is also small-scale. We only test graphs $n \leq 16$ while comparing one-hop local NBP against all-node global NBP. This does expose the leader-election scope bottleneck (Appendix B) but is not enough to characterize how belief scope, graph diameter and communication trade off at larger scale. A natural next step would be to sweep graph size and topology thoroughly and test intermediate r -hop and compressed belief summaries.

Although NBP addresses perception failures by making neighbour belief explicit, it’s not a general solution to tackle coordination. Future work should combine NBP with various mechanisms of symmetry breaking like private randomness, persistent priorities, and test whether these mechanisms repair structurally hard tasks without harming tasks where NBP already helps.

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A EXTENDED FAILURE ANALYSIS: METHODOLOGY AND TRANSCRIPTS

This appendix expands the failure analysis of §4. All material is from the `qwen3.5-4b` structured-output runs (the baseline cohort); consensus uses hard initialisation ($p=0.5$). For each run we have, per round, every agent’s declared state (the field `round_states`) and the full text of the messages it sent. We treat an agent’s own state as ground truth and ask how well the *claims* in its outgoing messages line up with it.

Claim extraction. On consensus the state vocabulary is binary, so claims are easy to parse reliably. We scan each outgoing message for three patterns and score each against the ground-truth `round_states` of the same round:

- **Own-state claim** — e.g. “my value is X”; matched against the sender’s own state.
- **Neighbour-state claim** — e.g. “you are at X”; matched against the recipient’s true state.
- **Agreement claim** — e.g. “consensus reached”, “finalising on X”; scored as *false* if the network is not actually unanimous that round. We count only declarations that agreement *has* been reached (or commit to a final value), not aspirations like “let’s reach consensus”.

A.1 CONSENSUS: CATALOGUE OF CLAIM FAILURES

Table 4 gives the aggregate counts behind Figure 5. The asymmetry is clearly in that the agents are almost perfect about their own state and barely above chance about a neighbour’s, and nearly half of all agreement claims are made while the network is split.

Table 4: Consensus claim accuracy (`qwen3.5-4b`, hard init $p=0.5$, pooled over all parsed messages). Chance for the binary state is 50%.

Claim type	Matched against	<i>n</i> claims	Correct / false
Own-state	sender’s own true state	132	98.5% correct
Neighbour-state	recipient’s true state	240	63.3% correct
Agreement	network actually unanimous	105	46.7% <i>false</i>

Own-state claims (the baseline competence). Agents reliably report their own value, confirming the failure is not in self-reporting:

```
[ws idx0, n=4] Marilyn, round 0 -- Marilyn’s true state: 0
Marilyn: "My state is 0. What is your state?" (correct)
[ws idx0, n=4] Jason, round 1 -- Jason’s true state: 0
Jason: "My current state is 0. Please consider aligning with 0." (
correct)
```

Neighbour-state claims (the blind spot). When an agent asserts a value *about* a neighbour it is wrong about a third of the time, often inventing agreement that does not exist:

```
[ws idx0, n=4] round 2 -- Marilyn’s true state: 0
Jason -> Marilyn: "I align with your 1 request." (Marilyn is
actually 0)
[ws idx0, n=4] round 2 -- Jason’s true state: 1
Sara -> Jason: "I hear you want to align with 0. Let’s stay with 0."
(Jason is at 1)
```

Agreement claims (false consensus). Misperception surfaces as confident but false declarations of agreement. The clearest cases are a lone dissenter who declares unanimity, and two coalitions that each announce “consensus” on opposite values in the *same* round:

```

[ba idx2, n=4] Lauren, round 4
  true states: Lauren=0, Evelyn=1, Alexander=1, Pamela=1
  Lauren: "Finalizing on 0 based on consensus from everyone." (Lauren
    is the only 0)

[ws idx0, n=4] round 2
  true states: Jason=1, Sara=0, Marilyn=0, Samantha=1
  Jason: "Let's finalize on 1 to reach consensus with Marilyn and
    Samantha."
  Marilyn: "Final decision: 0 - I agree with Jason and Sara."
    (two opposing 'consensus' declarations, network
    is 2-2 split)

```

A.2 LEADER ELECTION: BROKEN SYMMETRY BREAKING

A valid election requires *exactly one* Yes. Of 27 runs, only 15 succeed; 5 elect *zero* leaders and 7 elect *multiple* — i.e. 44% fail to break symmetry. The opening round is diagnostic: agents broadcast identity (“I am X”) rather than proposing a tie-break rule (e.g. largest name or degree), so several can settle on Yes simultaneously, or all defer to No.

```

MULTIPLE LEADERS [dt idx1, n=8] -- final answers contain two 'Yes' (
  Lauren, Brandon)
  Dorothy: "I am Dorothy. Initiating leader election process." state=
    No
  Lauren: "I am Lauren. Starting leader election process." state=
    No -> Yes
  Brandon: "I am Brandon" state=
    Yes
  (no agent ever references who else is claiming the role)

ZERO LEADERS [ws idx0, n=4] -- final answers all 'No'
  Samantha: "I am Samantha. I claim I am the leader candidate." state0=
    Yes
  (Samantha later abdicates to 'No'; nobody else steps up -> no leader)

```

A.3 COLORING: LOCKSTEP OSCILLATION

The signature coloring failure is synchronous motion: neighbours apply the same greedy update (“if I clash, move to the next group”) at the same time and therefore move together, so the clash never resolves. The run below is a complete graph on four agents (K_4 , all six edges present), which needs four distinct groups; instead three agents oscillate in perfect unison and the instance collapses to one group (hard pass = 0).

```

[ws idx2, n=4] K_4 (every pair adjacent) score = 0.00
  Janet : Group 1 -> Group 2 -> Group 3 -> Group 2 final = Group 1
  Kayla : Group 1 -> Group 2 -> Group 1 -> Group 2 final = Group 1
  Steven: Group 1 -> Group 2 -> Group 1 -> Group 2 final = Group 1
  Keith : Group 1 -> Group 2 -> Group 1 -> Group 2 final = Group 1

```

This is a synchrony error more than an information one. The agents already see their neighbours’ colours. It needs private randomness (or an asymmetric tie-break) to break, and NBP, that is a perception remedy, does not directly help here and slightly worsens it (§6.2).

A.4 MATCHING: UNRECIPROCATED COMMITMENT

A pair scores only if both agents name *each other*. The failure is the matching analogue of false consensus: an agent commits to a partner who has meanwhile committed elsewhere. In the run below, three agents converge on Jennifer or on already-None partners, leaving a chain of broken pairs.

```
[ba idx0, n=8]    score = 0.50
  final answers:  Helen->Vincent, Vincent->Helen    (the one good pair)
                  Jennifer->Amy    but Amy->None
                  Zachary->Jennifer, Stephen->Jennifer    but Jennifer->
                  Amy
                  Timothy->Sharon    but Sharon->None
```

Helen and Vincent reciprocate and succeed; everyone else commits to a partner with no maintained model of that partner’s current intention, and the selections never converge.

B LEADER ELECTION: THE SCOPE BOTTLENECK

Local NBP recovers part of the Structured-to-Global gap on leader election, concentrated at $n=16$ where diameters reach 5 (Figure 11). Local beliefs relay distant information through chains but do not re-emit it as durable entries for every node; Global NBP amplifies the elected leader’s identity across the network within a few rounds. At $n=16$, Global reaches 0.88 (soft) while Local plateaus at 0.50 — still well above Structured at 0.33. This is the one task where wider belief scope is worth its $O(n)$ cost, and it motivates studying intermediate k -hop scopes (we test only $k=1$ and $k=\infty$).

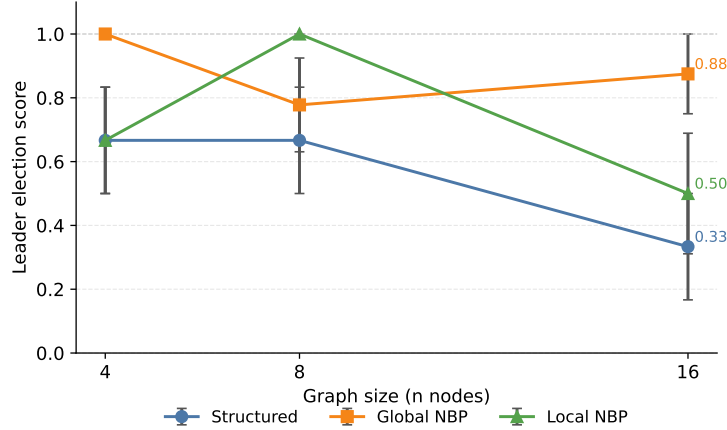


Figure 11: Leader-election soft score versus graph size (4B). Local NBP tracks Global through $n=8$ then plateaus at $n=16$, where Global retains its lift; both remain above Structured.

C CONSENSUS HARDNESS SWEEP: IS THE BENEFIT REAL?

We might argue that the NBP improvement on consensus is an artefact of a particular initialisation difficulty. To test this, we vary the initial probability p that each agent starts at ‘0’ from $p=0.5$ (maximally hard — perfect balance) to $p=0.9$ (easy — strong majority toward the focal point). The prediction: if NBP’s benefit is genuine coordination, it should be largest at $p=0.5$, where the protocol must do the most work, and shrink as $p \rightarrow 0.9$, where focal-point lock-in makes the protocol irrelevant. This is also what we observe (Figure 12): the Structured–NBP gap is widest at $p=0.5$ and closes as initialisation becomes easier.

D IMPLEMENTATION AND REPRODUCIBILITY

This appendix records the implementation details needed to reproduce every number in the paper. The driver is a lightly modified fork of the upstream AgentsNet harness (Grötschla et al., 2025); our additions are the belief-protocol layer (§5), the hard-initialisation parameter for consensus (§3.1), and the orchestration around it. The protocol layer is implemented entirely in the prompt and the

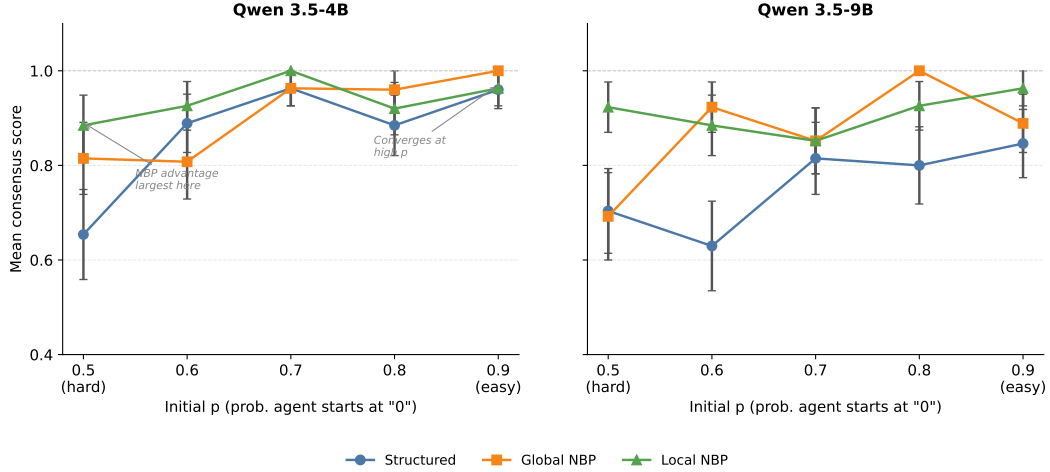


Figure 12: Consensus soft score versus initial-state probability p for the three protocols, both models. NBP’s advantage over Structured is largest at the hard end ($p=0.5$) and vanishes as $p \rightarrow 0.9$, confirming the gain reflects genuine coordination rather than an initialisation artefact.

response schema — no task-specific belief-update rule is hard-coded, and the same code path serves all five tasks.

Models and serving. We evaluate two open-weight models, qwen3.5-4b and qwen3.5-9b (Qwen Team, 2026), each served locally through vLLM (Kwon et al., 2023) behind an OpenAI-compatible API and reached via langchain-openai. The server runs a single model instance (`--tensor-parallel-size 1`, `--dtype auto`, `--max-model-len 131072`) with the Qwen3 reasoning parser enabled, on one A100 GPU. Each generation request is capped at `max_tokens=32768` to leave room for chain-of-thought, which is left enabled (the harness default). Client-side calls are throttled by an in-memory rate limiter (10 requests/s, bucket size 20) and wrapped in up to 10 retries on transient errors.

Decoding. We use the model’s default sampling configuration (no custom temperature is set in the harness); the only generation override is the token cap above. The reasoning trace is parsed off and discarded by the server’s Qwen3 reasoning parser, so only the final JSON object reaches the harness. The server was configured for Qwen3 parsing, but malformed outputs still occurred and were handled as parse failures.

Graphs and sweep design. All graphs are loaded from the public HuggingFace dataset disco-eth/AgentsNet (Grötschla et al., 2025), keyed by `(graph_generator, num_nodes, index)`. We sweep three generators — Watts–Strogatz (`ws`), Barabási–Albert (`ba`), and Delaunay (`dt`) — at graph sizes $n \in \{4, 8, 16\}$, with 3 samples per generator. This gives 9 graphs per `(task, n)` cell and 27 graphs per `(task, protocol, model)` cell, the unit behind every headline mean. A fixed seed (1729) controls sample selection and the consensus initialisation draw, so reruns are deterministic.

Round schedule. Following AgentsNet, the number of message-passing rounds is forced to $R = 2 \cdot \text{diameter}(G) + 1$ for consensus and leader election (and for any graph with $n > 16$); any `--rounds` flag is ignored in those cases. For matching, coloring, and vertex cover at $n \leq 16$ the round budget is set explicitly by graph size: $R = 4$ at $n=4$, $R = 5$ at $n=8$, and $R = 6$ at $n=16$.

Structured outputs and parse robustness. Both per-round messages and final answers are constrained with a JSON-schema `response_format`. The round schema requires a `state` field, a `messages` object keyed only by valid recipients, and — under NBP — a `belief_state` object (one typed entry per agent in scope; §5). The

final-answer schema constrains final_answer to the task’s valid options. Parsing is defensive: a round response that fails to parse triggers one re-prompt, and repeated failures are counted (num_fallbacks, num_failed_json_parsings_after_retry, num_failed_answer_parsings_after_retry) and stored with the run. The belief_state block is parsed tolerantl such that a missing or malformed belief entry is dropped rather than failing the round.

Consensus initialisation. In the upstream framing, consensus agents receive no initial state and choose 0/1 freely, which (§3.1) lets the model’s prior toward 0 manufacture a focal-point head start. Our hard-init mode assigns each agent an i.i.d. Bernoulli($1-p$) start over $\{0, 1\}$, where p is the target fraction at 0; $p=0.5$ is maximally hard (balanced) and is the setting for all primary consensus rows (cohorts B–D). The hardness sweep (§C) varies p from 0.5 to 0.9. The realised initial_states, random_init_enabled, and random_init_p are saved with every run.

Cohorts and how to reproduce each. The four cohorts of Table 2 map onto the harness as follows. *Structured* (A) is the default structured-JSON path (`--belief_mode none`). *Global NBP* (B) adds the network-wide belief vector (`--belief_mode b3`, $O(n)$ entries per message). *Local NBP* (C) restricts the vector to self plus direct neighbours (`--belief_mode b3-neighbours-only`, $O(k)$ entries); distant information still reaches an agent through the beliefs its neighbours relay, but is not re-emitted in its own schema. Consensus runs in all NBP/Structured cohorts add `--random_init --random_init_p 0.5`.

Outputs. Each run writes one JSON file under `{output_dir}/{model}/{task}/`, recording the full per-agent transcripts, per-round round_states, per-round belief_round_states (NBP only), final answers, score, the graph, the git commit hash, the fallback counters, and the initialisation record. The notebooks under notebooks/ (paper_narrative.ipynb for tables/figures, qualitative_analysis.ipynb for the claim and belief-calibration analyses) regenerate every figure and table in the paper from these files. The cohort of a run is inferred from its output directory (e.g. b3-neighbour $\rightarrow$ Local NBP, b3 $\rightarrow$ Global NBP, structured-output $\rightarrow$ Structured), and duplicate reruns of the same graph instance are deduplicated by keeping the latest file.

E PROMPTS

This appendix gives the verbatim prompt scaffolding. The base system prompt below is shared by all cohorts; the NBP cohorts append one of the two belief blocks; every cohort closes each instance with the same final-answer prompt. Task-specific text (short_task_description, long_problem_description, question, and the valid answer set) is supplied by the task class and substituted into the placeholders.

Base system prompt. Sent once at bootstrap to every agent; {num_agents}, {own_name}, {neighbors}, and {num_rounds} are filled per node.

```
You are an agent that is connected with other agents (your neighbors),
  who you communicate with. Your neighbors can in turn communicate with
  their neighbors and so forth. {short_task_description}
The rules are as follows:
1. There are {num_agents} agents in total. Everybody has a unique name.
   Your name is {own_name}.
2. You can only communicate with your immediate neighbors ({neighbors}).
   You cannot see or directly communicate with anyone else, unless
   information is relayed by intermediate agents.
3. You can exchange text-based messages with your neighbors in rounds.
   In each round, you will first receive the last messages sent by your
   neighbors and then be asked to generate your response messages which
   your neighbors receive in the next round. This process repeats for {
   num_rounds} rounds of message passing. Importantly, the process is
   synchronous: Every agent decides on which messages to send at the
   same time and sees the messages from other agents only in the next
   round.
```

4. Everybody (including you) decides what to share or request from neighbors. In every round, decide your current task state and the next set of messages you want to send. The "state" field must be your current tentative answer using one of the valid final answer options for the task. Output only a JSON object with your state and response messages.
 5. The messages you send to your neighbors are formatted under the "messages" JSON key. For example, if your neighbors are Alan and Bob, your output should look as follows:


```
{
  "state": "Your current tentative answer.",
  "messages": {
    "Alan": "Message that will be sent to Alan.",
    "Bob": "Message that will be sent to Bob."
  }
}
```
- It is not mandatory to send a message to every neighbor in every round. If you do not want to send a message to a particular neighbor, you may omit their name from the JSON.
6. After {num_rounds} message passes, you have to solve the following task: {long_problem_description}

Belief-state instructions — Global NBP. Appended to the base prompt under `--belief_mode b3`. Each message must additionally carry a `belief_state` entry for every agent in the network.

BELIEF STATE INSTRUCTIONS:
 Every round, your JSON output must include an additional "belief_state" field alongside "state" and "messages". This is a structured record of what you believe about every agent's current state in the network. Each entry maps an agent name to {"state": <one of the valid task answers, or "unknown">, "confidence": "high" | "medium" | "low" | "unknown"}.

Cover every agent in the network (including yourself).

Update belief_state each round from (1) messages your direct neighbours sent you and (2) the belief states they relayed about agents you cannot observe directly. Confidence levels:

- "high" : observed directly this round
- "medium" : relayed one hop
- "low" : two or more hops, or stale
- "unknown": no information

Then use your own belief_state to reason about the state of the network as a whole, and choose your "state" field in light of that picture.

Example output shape:

```
{
  "state": "<your tentative answer>",
  "belief_state": {
    "<your-own-name>": {"state": "...", "confidence": "high"},
    "<some-neighbour>": {"state": "...", "confidence": "high"},
    "<distant-agent>": {"state": "...", "confidence": "low"}
  },
  "messages": {
    "<some-neighbour>": "Message text..."
  }
}
```

Belief-state instructions — Local NBP. Appended under `--belief_mode b3-neighbours-only`. Identical in spirit but the belief vector is restricted to self plus direct neighbours; distant information is consumed from relayed beliefs but never re-emitted.

BELIEF STATE INSTRUCTIONS:

Every round, your JSON output must include an additional "belief_state" field alongside "state" and "messages". This is a structured record of what you believe about yourself and your direct neighbours only. Each entry maps an agent name to {"state": <one of the valid task answers, or "unknown">, "confidence": "high" | "medium" | "low" | "unknown"}.

Cover yourself and your direct neighbours only. Information about more distant agents reaches you via the belief states your neighbours relay; use that information to inform your own "state" choice, but do NOT add belief entries for distant agents to your output.

Update belief_state each round from (1) messages your direct neighbours sent you and (2) the belief states they relayed about agents you cannot observe directly. Confidence levels:

- "high" : observed directly this round
- "medium" : relayed one hop
- "low" : two or more hops, or stale
- "unknown": no information

Then use your own belief_state (and the relayed beliefs in messages) to reason about the state of the network as a whole, and choose your "state" field in light of that picture.

Example output shape (assuming your neighbours are Alan and Bob):

```
{
  "state": "<your tentative answer>",
  "belief_state": {
    "<your-own-name>": {"state": "...", "confidence": "high"},
    "Alan": {"state": "...", "confidence": "high"},
    "Bob": {"state": "...", "confidence": "medium"}
  },
  "messages": {
    "Alan": "Message text..."
  }
}
```

First-message and final-answer prompts. After the system prompt, round 0 is elicited with a fixed bootstrap turn, and after the last round each agent is asked for its committed answer, constrained to the task's valid options.

[bootstrap, round 0]

What are the first messages you want to send to your neighbors? Output only JSON in the format specified earlier.

[final answer]

{question} Output only a JSON object with exactly one key, "final_answer". The value must be one of these valid options: {valid_answers}

On receipt, the harness strips the recipient's own entry out of each sender's belief_state before relaying it, so an agent never reads back its own belief about itself; everything else in the message is passed through unchanged.